Shaashvat Shetty

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Education

B.S. Computer Science | B.S. Statistics Expected Graduation: December 2026
University of Maryland, College Park

GPA: 3.800 (CS Honors, Dean's List)

Relevant Coursework: Advanced Data Structures, Algorithms, Data Science & Machine Learning, Computer Systems, Statistical Computing with SAS, Applied Probability & Statistics

Experience

CRA-WP DREU Research Intern – Multimodal LLM Representation Analysis Summer 2026
University of Maryland, College Park

- Researching fine-grained KV cache sparsity in open-source vision-language models to analyze how image and text tokens are represented during inference.
- Profiling Key/Value cache tensor magnitude distributions across layers to compare visual-token and text-token behavior, including modality-specific outlier patterns.
- Implementing Python-level modality-aware pruning logic with separate sparsity targets for image-token and text-token KV caches using an existing CUDA backend.
- Evaluating uniform vs. modality-aware pruning on TextVQA/MMMU-style benchmarks to study accuracy, memory, and robustness trade-offs.

Research Assistant August 2025 – Present
University of Maryland, College Park — Advisor: Prof. Haizhao Yang

- Studying how AI can accelerate materials discovery by generating interpretable mathematical formulas instead of relying on black-box prediction models.
- Comparing Monte Carlo Tree Search and LLM-based symbolic regression as search strategies for discovering candidate structure-property relationships.
- Implementing experimental pipelines to generate, score, and analyze symbolic expressions across accuracy, complexity, and search efficiency.
- Co-authoring a research paper evaluating when LLM-guided search improves symbolic discovery compared with traditional tree-search methods.

Software Engineering Intern June 2024 – September 2024
Juniper Networks, Sunnyvale, CA

- Built a Python-based network automation CLI for CLOS architectures to validate EVPN-VXLAN connectivity, reducing troubleshooting from hours to under 10 seconds.
- Implemented a second CLI command to explain failures by identifying likely misconfigurations such as VLAN mismatches and interface issues.
- Validated workflows across multi-leaf/spine topologies with hundreds of nodes and wrote tests/documentation to support team handoff.

Computer Vision Intern June 2022 – August 2022
UCSC SIP, Santa Cruz, CA

- Built an ML system for drone navigation from natural-language commands by combining object detection with text-to-object matching.
- Evaluated YOLOv5 vs. YOLOv3 and selected YOLOv3 for ∼35% faster inference on constrained hardware.
- Improved robustness to synonym variations in commands using cosine similarity without retraining.

Technical Skills

Languages: Python, C, Bash, SQL, Java, JavaScript, HTML/CSS, R, Rust, OCaml, SAS, MATLAB, Go
ML/Research Tools: PyTorch, Hugging Face, NumPy, Pandas, scikit-learn, Matplotlib, Jupyter Notebook, TensorFlow, Keras, YOLOv3
Systems/Tools:Git, GitHub, VS Code, JUnit, GPU Memory Optimization, LLM Inference, KV Cache Analysis, EVPN-VXLAN, BGP, Data Center Tech, Raspberry Pi

Projects

HeyDoug — Multilingual Code Tutor (VS Code Extension) *April 2026*

- Built a VS Code extension that analyzes multi-file codebases, highlights bugs, and suggests fixes in real time via LLM.
- Enabled voice-based interaction where users can speak to the system and receive guided explanations of code logic via speech-to-text.
- Deployed lightweight backend on Raspberry Pi for local, low-latency inference.

Custom iMessage Chatbot *July 2025*

- Parsed and structured 50k+ personal iMessages into JSONL format for model fine-tuning.
- Fine-tuned TinyLlama-1.1B via Hugging Face to learn and replicate a personal conversational style.
- Developed a lightweight HTML/CSS/JS interface to query the model and generate tone-matched replies.

Fashion MNIST Classification Analysis *May 2025*

- Benchmarked 7 classification models using PCA, 5-fold cross-validation, confusion matrices, and performance trade-off analysis.
- Engineered a CNN that achieved 91.05% accuracy, outperforming traditional machine learning baselines.